

MUKESH KUMAR SHARMA

Staff / Lead iOS Engineer · AI-Native Mobile · Swift · SwiftUI · LLM Streaming · Foundation Models · On-Device LoRA

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SUMMARY

Lead iOS engineer with **14+ years** shipping to the App Store — **10+ years in Swift**, 5+ in SwiftUI — now building **AI-native mobile experiences**. Architected Vanguard's generative-AI investing assistant end to end: **real-time LLM response streaming**, an actor-isolated tokenizer, and a reusable SwiftUI component library adopted by other engineers. Independently built an on-device AI product that **fine-tunes Apple Foundation Models with LoRA adapters** for live two-party video-consultation transcription, including its **Node.js/TypeScript backend**. Published one of the first native Swift OpenAI clients (Jan 2023). Canadian citizen — **ready to relocate immediately** to San Francisco, New York, or Seattle.

SELECTED HIGHLIGHTS

- **Modern ML/AI in shipped apps:** real-time LLM streaming chat in a flagship wealth-management iOS app; **LoRA-fine-tuned Apple Foundation Models (iOS 26/27)** running fully on device; *chatGPT_iOS* (Jan 2023), an early public native Swift OpenAI client, plus on-device Stable Diffusion — both open source.
- **0→1 product building:** founded and solo-shipped an on-device AI healthcare product to production pilot; led 0→1 delivery of an enterprise payments platform; shipped a smart-city app's first production release.
- **High-adoption apps:** features live for the full Kroger customer base, millions of TD Canada banking customers, and GAP / Athleta / Banana Republic / Old Navy.
- **Full-stack & platform depth:** Node.js + TypeScript backends on Google App Engine; audio/video streaming (Twilio Video, AVFoundation, Apple Speech); frameworks adopted org-wide — NEOMCore SPM (~20 apps), payments SPM (3 apps), org-wide CI/CD.

TECHNICAL SKILLS

Languages & UI: Swift (6 / 5.x), SwiftUI, UIKit (programmatic + storyboard), Objective-C, TypeScript/JavaScript · latest platform APIs: Observation/@Observable, TipKit, AppIntents, WidgetKit · Auto Layout, Core Animation, Core Graphics, MapKit · **Audio/video & streaming:** AVFoundation, Twilio Video, Apple Speech (real-time transcription), text-to-speech

Concurrency & Reactive: async/await, Actors, custom `@globalActor`, Sendable, structured concurrency, strict concurrency checking · Combine (PassthroughSubject, stream subscription), Observation / `@observable`, GCD

Architecture: MVVM, MVVM + Coordinator, VIP/Clean, SOLID, protocol-oriented design, dependency injection, repository pattern, SPM modularization (25 internal packages), Apple HIG

On-Device & Applied AI: Apple Foundation Models framework (iOS 26), Core ML, LoRA adapter-based on-device fine-tuning, OpenAI SDK, real-time response streaming (SSE, newline-delimited JSON), custom stream tokenizers, prompt engineering, LangChain prompt templates, on-device Speech-to-Text, on-device Stable Diffusion (no server round-trip)

Networking, Data & Security: REST/JSON, URLSession, Alamofire (DataStreamRequest), GraphQL (Apollo iOS), WebSockets, Codable · Core Data, Firebase (Auth, Firestore, FCM) · Keychain, LocalAuthentication (Face ID/Touch ID), OAuth 2.0, SSL pinning, App Transport Security, APNs, Universal Links

Testing & Delivery: XCTest, Swift Testing, XCUITest, TDD, mocks/stubs · Fastlane (Ruby), GitHub Actions, Xcode Cloud, TestFlight & App Store release management · Instruments (Time Profiler, Allocations, Leaks), Crashlytics, Sentry

Cross-Platform & Backend: React Native (iOS + Android, production) with native Swift↔JS module bridging · Node.js + TypeScript REST APIs on Google App Engine

Quality & Process: Accessibility (VoiceOver, Dynamic Type, Accessibility Inspector audits) · Firebase Remote Config, feature flags, A/B testing, Grafana · Agile/Scrum, design reviews, technical documentation, mentoring · JIRA, Confluence, Bitbucket

PROFESSIONAL EXPERIENCE

Senior iOS Engineer — Vanguard (contract via Publicis Sapient)

Mar 2026 – Aug 2026

Explore & Empower — generative-AI investing assistant in Vanguard's flagship iOS app · Toronto (hybrid)

- Delivered the AI assistant **end to end in Swift and SwiftUI across 8 modules**, including a **50-file SwiftUI chat component package** — shipped to the client's **phase 1 production release** at a 0.1% customer rollout.
- **Co-designed the streaming response contract with the backend and AI-platform teams**, then implemented **real-time LLM streaming** over Alamofire DataStreamRequest with a custom newline-delimited JSON tokenizer, publishing decoded chunks to the UI via a Combine PassthroughSubject as they arrive.
- Adopted **Swift Concurrency end to end** — an actor-isolated tokenizer, a custom `@globalActor` for serialized stream processing, `@MainActor` UI boundaries, and full Sendable conformance — eliminating data races under load.
- Drove the **component-discriminated streaming schema to agreement across iOS, backend, and design**: 8 renderable response types (markdown, tables, lists, charts) with custom Decodable initializers, parse-boundary validation, and graceful degradation.
- Shipped a **reusable coach-mark framework (17 files)** modeled on TipKit but back-deployed to iOS 17, plus a 5-page animated onboarding flow — **adopted by other engineers**; validated through iOS 26.
- Modularized the feature into **internal SPM packages** with a protocol + paired mock per ViewModel over a repository pattern, keeping it independently testable and releasable from the host app; covered tokenizer, DTO decoding, and ViewModels with XCTest and Swift Testing.
- Authored **custom Claude AI development commands** that automate first-pass code review and enforce the team's Swift/SwiftUI coding guidelines across the feature.

Lead iOS Engineer — Tech Mahindra

Dec 2024 – Mar 2026

Concurrent client delivery: TruPOS — enterprise payments platform (iOS) · TruGreen — consumer app (React Native, iOS + Android)

- Led **0→1 delivery** of the enterprise's central payment iOS app in Swift/SwiftUI/Combine — the single checkout hub for multiple business lines — owning MVVM architecture, checkout, transaction history, and merchant onboarding.
- Designed an **offline transaction queue** (Core Data + BGTaskScheduler, 72-hour retention) letting field sales take payments offline and auto-sync on reconnect — **reducing offline transaction failures to effectively zero**.
- Built a reusable **Swift Package for payments, receipts, and deep linking adopted across 3 internal apps**, eliminating per-app reimplementation.
- Owned **CI/CD via Fastlane, GitHub Actions, and Xcode Cloud** — automated TestFlight and App Store releases, replacing a manual process and cutting release cycle time **20–50%**.
- Integrated **Bluetooth card readers and MFi accessories** with background-mode support and a custom retry/error layer for intermittent connectivity.
- Achieved **80% unit-test coverage** on business logic (XCTest/XCUITest); profiled with Instruments to fix retain cycles and improve cold-start and memory pressure.
- **Partnered with backend engineers across payment, authentication, and observability** to align REST contracts end to end, and ran **design reviews with product owners, designers, and QA** — driving data-requirement and risk-mitigation documentation for the payment platform.
- In parallel on **TruGreen**, shipped consumer features in **React Native (TypeScript) to both App Store and Play Store** from one codebase, and authored **native iOS modules in Swift bridged to the JS layer** (biometrics, deep links, secure storage) that pure JS could not reach — debugging across Metro, Hermes, and the native bridge.

Founder & Lead Engineer — Doctor AI (independent product)

Jan 2024 – Present

On-device AI healthcare product — iOS + Android + Node.js backend · limited production pilot with early customers

- **Fine-tuned Apple Foundation Models with LoRA adapters** on iOS 26/27 for a domain-specific medical-consultation use case — specializing clinical vocabulary and note summarization, with **tuned system instructions and default safety guardrails** layered over the adapter — inference runs **entirely on device**, keeping patient audio and transcripts off any server.
- Built **live two-party transcription for video consultations** — real-time speech-to-text for both doctor and patient during Twilio Video calls, plus text-to-speech playback — on Apple Speech and the LoRA-tuned on-device model.
- Built and shipped the entire stack solo across **6+ repositories** — iOS client, Android companion, shared Swift Package, and a **Node.js + TypeScript backend on Google App Engine** (REST, auth, sessions, LLM orchestration); owned API contracts, schema design, deployment, and release management end to end.

Senior / Lead iOS Engineer — Tech Mahindra (NEOM Smart City, KSA)

Jun 2023 – Jan 2024

- Architected and **shipped the first production App Store release** of the NEOM Smart City app for residents, visitors, and service providers.
- Defined the architecture, interface contracts, and versioning strategy for **NEOMCore — a reusable SPM framework** (networking, OAuth, SSL pinning, shared UI) **adopted across ~20 internal NEOM apps**; integrated Apollo iOS for type-safe GraphQL.
- Integrated the **OpenAI iOS client and a LangChain-based prompt SDK** to power an in-app digital assistant for utilities, transport, and support; added Twilio audio/video for resident-to-agent support and Apple Speech for voice interaction.
- **Led a small iOS team** through the full lifecycle — ran design reviews with product, backend, and security, owned architectural decisions, and **mentored 3 engineers** to a production launch.

Senior / Lead iOS & Release Engineer — Tech Mahindra (Kroger & banner apps)

Jan 2023 – Jun 2023

- Owned end-to-end iOS delivery of the **Kroger Gamification module (v1.0 and v2.0)**, released to the **full customer base** across Kroger, Ralph's, Fred Meyer, and Harris Teeter.
- Authored the **DevOps pipelines adopted by the entire iOS org** across Kroger and its banner teams (Fastlane, GitHub Actions, shell/Ruby/Node.js) for build, sign, and deploy; instrumented rollouts with Firebase Remote Config flags and A/B experiments monitored via Crashlytics crash-free rate and Grafana.

Senior iOS Engineer — Avenue Code @ GAP Inc.

Jun 2022 – Dec 2022

- Delivered features across **GAP, Athleta, Banana Republic, and Old Navy** storefronts in Swift/SwiftUI and Objective-C; led the Notification System workstream from scratch to production launch with product and backend partners.

Senior iOS Engineer — Clearbridge Mobile / Amdocs (TD Canada banking app)

Sep 2021 – Jun 2022

- Led development of the **SIJ, MHA, and CIBA loan modules** in Swift and Objective-C, shipping features used by **millions of TD banking customers**.
- Integrated **Face ID / Touch ID via LocalAuthentication and Keychain** for secure authentication and hardened the app against jailbreak and tampering; contributed to the **TD iOS Banking Foundation Framework**, republished via CocoaPods.

Lead iOS Engineer — Tata Consultancy Services (Trickl Utility & Green Button)

Nov 2017 – Aug 2021

- Owned the iOS codebase for a smart-utility app (real-time energy gauges, device control, push) over an AWS backend, and built cross-platform Green Button apps for **Toronto, London, and Whitby Hydro**.

iOS Engineer — Tata Consultancy Services (Pfizer) · **GlobalLogic, smartData, Regard Solutions**

Aug 2011 – Nov 2017

- Shipped **six internal Pfizer iOS apps** (Objective-C/Swift — custom Core Graphics charts, Janrain SSO, offline sync) and **8+ App Store apps** across healthcare, retail, and consumer; contributed accessibility R&D for the Qualcomm Healthy Circle POC.

OPEN SOURCE

Open-Source iOS + AI — github.com/imuks

2023 – Present

- **chatGPT_iOS** — Swift OpenAI client, streaming + SwiftUI (Jan 2023). **StableDiffusion_iOS** — on-device generation. **swift_DSA** — Swift DSA reference.

EDUCATION B.E., Computer Technology — Priyadarshini College of Engineering, R.T.M. Nagpur University, 2011